

Mathematics for Computer Science – CRT

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June 8, 2026

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思路.

- 数论部分，目前我们学习解方程方法包括：一元一次同余方程，一元 n 次同余方程。接下来需要扩展到一元同余方程组的解法。未来是多元同余方程等等。
- 群论部分，主要目标是群结构。群结构与解同余方程有什么关系？

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目标

给出一种求解一元一次同余方程组的高效算法；探讨群结构与解同余方程组的关系。

Chinese Remainder Theorem (中国剩余定理), 或称为中国余数定理则更准确。

历史.

中国剩余定理有着满满的中国传统文化元素, 与之相关的名人包括: 春秋时期的孙子、秦汉时期的韩信、宋朝的沈括和秦久韶等。南北朝时期(公元 5 世纪)的数学著作《孙子算经》卷下第二十六题所谓“物不知数”问题则为国人所熟知: 有物不知其数, 三三数之剩二, 五五数之剩三, 七七数之剩二。问物几何?

中国剩余定理讨论一元同余方程组的高效解法。

Example (Example 1.)

Suppose we wish to solve:

$$x \equiv 2 \pmod{5}$$

$$x \equiv 3 \pmod{7}$$

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Suppose we wish to solve:

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Solution.

1. We have $x = 5t + 2$ from the first congruence, for $t \in \mathbb{Z}$;

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3. Simplifying, we get $5t \equiv 1 \pmod{7}$;
4. Multiplying both sides by 5^{-1} , we get $t = 7s + 3$ for $s \in \mathbb{Z}$;

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4. Multiplying both sides by 5^{-1} , we get $t = 7s + 3$ for $s \in \mathbb{Z}$;
5. Finally, $x = 35s + 17$, which means $x \equiv 17 \pmod{35}$.

思考.

以上求解一元同余方程组的方法是否高效？如果不高效，能否给出更高效的算法？

The Chinese Remainder Theorem–CRT.

For any system of equations like this, the *Chinese Remainder Theorem*, short for CRT, tells us there is always a unique solution up to a certain modulus, and describes how to find the solution efficiently.

Theorem

Let p, q be primes, $n = pq$. For each $a \in \mathbb{Z}_p$, $b \in \mathbb{Z}_q$, there is a unique x , $0 \leq x < n$ such that $x \equiv a \pmod{p}$ and $x \equiv b \pmod{q}$.

The Chinese Remainder Theorem–CRT.

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Let p, q be two relatively prime positive integers, $n = pq$. For each $a \in \mathbb{Z}_p$, $b \in \mathbb{Z}_q$, there is a unique x , $0 \leq x < n$ such that $x \equiv a \pmod{p}$ and $x \equiv b \pmod{q}$.

Proof Idea

- 1 Given a and p , how can we find some c s.t. $ac \equiv a \pmod{p}$?

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- 1 Given a and p , how can we find some c s.t. $ac \equiv a \pmod{p}$?
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- 1 Given a and p , how can we find some c s.t. $ac \equiv a \pmod{p}$?
- 2 c must be some 1 under modulo p
- 3 模 p 下为 1 的数可能有哪些 (可参考 CINTA 附录 D.) ?
- 4 Find some c s.t. $acc^{-1} \equiv a \pmod{p}$
- 5 Then x must be something like that $x = (acc^{-1} + bdd^{-1})$, what should be c and d ?

The Chinese Remainder Theorem–CRT.

Theorem

Let p, q be coprime positive integers, $n = pq$. For each $a \in \mathbb{Z}_p$, $b \in \mathbb{Z}_q$, there is a unique x , $0 \leq x < n$ such that $x \equiv a \pmod{p}$ and $x \equiv b \pmod{q}$.

Proof.

By construction. Since p, q are coprime, there exist p_1 and q_1 such that $p_1 \equiv p^{-1} \pmod{q}$ and $q_1 \equiv q^{-1} \pmod{p}$. Define x by:

$$x = aqq_1 + bpp_1$$

It is easy to check that x satisfies both equations. It remains to show no other solutions exist modulo n . Suppose $\exists z \neq x$ is another solution. Then $(z - x) = tp$ and $(z - x) = sq$, for some $t, s \in \mathbb{Z}$. Since p and q are coprime, then $(z - x) = kpq$, for $k \in \mathbb{Z}$. Hence $z \equiv x \pmod{n}$.



Example.

Example (Example 2.)

Suppose we wish to solve:

$$x \equiv 2 \pmod{5}$$

$$x \equiv 3 \pmod{7}$$

Solution.

1. Let $a = 2$, $b = 3$, $p = 5$, $q = 7$, $n = pq = 35$;
2. Compute $p_1 \equiv p^{-1} \pmod{q}$ and $q_1 \equiv q^{-1} \pmod{p}$ using egcd algorithm; $p_1 = 3$, $q_1 = 3$;
3. $x \equiv aq_1 + b p p_1 \pmod{n}$; $x \equiv 17 \pmod{35}$;
4. It is easy to check that x is a correct solution.

求解以下方程.

Suppose we wish to solve:

$$x \equiv 3 \pmod{11}$$

$$x \equiv 4 \pmod{13}$$

Exercise.

求解以下方程.

Suppose we wish to solve:

$$x \equiv 3 \pmod{11}$$

$$x \equiv 4 \pmod{13}$$

Ans.

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For several equations, we have a generalized version of CRT.

Theorem

Let $m_1, m_2, \dots, m_n$ be a set of pairwise relatively prime positive integers. Then the system of n equations:

$$x \equiv a_1 \pmod{m_1}$$

$\dots$

$$x \equiv a_n \pmod{m_n}$$

has a unique solution for x modulo M where $M = m_1 m_2 \dots m_n$.

Generalization.

Proof.

By construction. Let $M = \prod_{i=1}^n m_i$, $b_i = M/m_i$, $b'_i = b_i^{-1} \pmod{m_i}$. Then

$$y = \sum_{i=1}^n a_i b_i b'_i \pmod{M}$$

is the unique solution. □

思考题-1.

思考题.

设 m 和 n 为互素的正整数, $a > 0$ 为一个正整数, 如果

$$x \equiv a \pmod{m}$$

$$x \equiv a \pmod{n}$$

x 模 mn 等于什么? 为什么?

思考题-2.

如果同余方程组的模数 $m_1, m_2, \dots, m_n$ 不两两互质, 那么上述方程组是否还有解? 如果有解, 怎么求? 请思考下面这题。

思考题.

判定下列两个同余方程组是否有解。若有, 求出模适当模数下的所有解。

$$(a) \begin{cases} x \equiv 7 \pmod{12} \\ x \equiv 3 \pmod{8} \end{cases}$$

$$(b) \begin{cases} x \equiv 5 \pmod{12} \\ x \equiv 2 \pmod{8} \end{cases}$$

思考题-2.

如果同余方程组的模数 $m_1, m_2, \dots, m_n$ 不两两互质，那么上述方程组是否还有解？如果有解，怎么求？请思考下面这题。

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思路

模数不互素，也就是说有大于 1 的公共因子 d ，这个 d 能起到什么作用？其次，反向应用思考题-1 的结论，从一个方程式化为多个方程式。

A perspective from Abstract Algebra.

提醒：本节关注的不是算法，而是 CRT 背后的思想。

Motivation.

Let $n = pq$, where $p, q > 1$ are relatively prime. Given a positive integer x , it can be expressed as a unique pair $([x \bmod p], [x \bmod q])$.

A perspective from Abstract Algebra.

Theorem

Let $p, q > 1$ be coprime, $n = pq$. Then

$$\mathbb{Z}_n \cong \mathbb{Z}_p \times \mathbb{Z}_q \quad (\text{as an additive group})$$

and,

$$\mathbb{Z}_n^* \cong \mathbb{Z}_p^* \times \mathbb{Z}_q^* \quad (\text{as a multiplicative group}).$$

Proof.

1. Define $\phi : \mathbb{Z}_n \rightarrow \mathbb{Z}_p \times \mathbb{Z}_q$ by:

$$\phi(x) \triangleq ([x \bmod p], [x \bmod q])$$

2. Show ϕ is bijective.

3. Check that $\phi(x)$ preserves the group operation.

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3. Check that $\phi(x)$ preserves the group operation.

The proof that it is an isomorphism from $\mathbb{Z}_n^*$ to $\mathbb{Z}_p^* \times \mathbb{Z}_q^*$ is similar.



Remark

首先，注意到 $\mathbb{Z}_p \times \mathbb{Z}_q$ 是加法群，请验证！

How to prove the bijection?

Proof.

证明映射 ϕ 是一种双射，即证明 ϕ 是满射且单射。满射显然，因为根据中国剩余定理，任意序对中的两个同余式在模 n 下存在唯一解。证明单射即证明，如果对任意整数 $0 \leq a, b < n$ ，有 $([a \bmod p], [a \bmod q]) = ([b \bmod p], [b \bmod q])$ ，则 $a = b$ 。再次根据中国剩余定理可得。 □

How to prove the isomorphism?

Proof.

证明映射 ϕ 保持群操作，即需要证明：

$$\begin{aligned}\phi(a + b) &= ([(a + b) \bmod p], [(a + b) \bmod q]) \\ &= ([a \bmod p], [a \bmod q]) + ([b \bmod p], [b \bmod q]) \\ &= \phi(a) + \phi(b)\end{aligned}$$



Example.

Example (Example 3.)

Take $n = 15 = 5 \cdot 3$. $\mathbb{Z}_n^* = \{1, 2, 4, 7, 8, 11, 13, 14\}$ is isomorphic to $\mathbb{Z}_5^* \times \mathbb{Z}_3^*$ since we can give the following correspondence:

$$1 \leftrightarrow (1, 1) \quad 2 \leftrightarrow (2, 2) \quad 4 \leftrightarrow (4, 1) \quad 7 \leftrightarrow (2, 1)$$

$$8 \leftrightarrow (3, 2) \quad 11 \leftrightarrow (1, 2) \quad 13 \leftrightarrow (3, 1) \quad 14 \leftrightarrow (4, 2)$$

Example.

Example (Example 4.)

To compute $14 \cdot 13 \bmod 15$. Since $14 \leftrightarrow (4, 2)$ and $13 \leftrightarrow (3, 1)$, we have:

$$(4, 2) \cdot (3, 1) = ([4 \cdot 3 \bmod 5], [2 \cdot 1 \bmod 3]) = (2, 2).$$

Note that $(2, 2) \leftrightarrow 2$, which is the correct answer.

Example.

Example (Example 5.)

To compute $11^{53} \bmod 15$. Since $11 \leftrightarrow (1, 2)$ and $2 \equiv -1 \bmod 3$ we have:

$$(1, 2)^{53} = ([1^{53} \bmod 5], [(-1)^{53} \bmod 3]) = (1, -1 \bmod 3) = (1, 2).$$

Thus, $11^{53} \bmod 15 = 11$

Example (Example 6.)

设 $n = p \cdot q$ 为合数, p 和 q 是两个不同的奇素数。以下等式在模 n 的意义上有多少个解?

$$x^2 \equiv 1 \pmod{n}$$

为此, 需要解以下两个同余方程

$$x^2 \equiv 1 \pmod{p}$$

$$x^2 \equiv 1 \pmod{q}$$

以上两式分别有两个不同的解。因此在模 n 的意义上总共有 4 个解。同时, 这也就解决了上一章的一个遗留问题。

Example.

Example (Example 7.)

To compute $12^{12} \bmod 143$

Example.

Example (Example 7.)

To compute $12^{12} \bmod 143$ 解:

- 记 $n = 143$, $n = p \cdot q = 13 \cdot 11$;
- 求: $12^{12} \bmod 13$ 和 $12^{12} \bmod 11$;
- 所以, $12^{12} \bmod 143 = 1$.

Little thought.

Remark

A practical application: if we have many computations to perform on $x \in \mathbb{Z}_n^$ (e.g. RSA signing and decryption), we can convert x to $(a, b) \in \mathbb{Z}_p^* \times \mathbb{Z}_q^*$ and do all the computations on a and b instead before converting back.*

This is often cheaper because for many algorithms, doubling the size of the input more than doubles the running time.

Homework.

Homework.

1. Use CRT to solve:

$$x \equiv 8 \pmod{11}$$

$$x \equiv 13 \pmod{19}$$

2. Use CRT to solve the system of congruences:

$$x \equiv 1 \pmod{5}$$

$$x \equiv 2 \pmod{7}$$

$$x \equiv 3 \pmod{9}$$

$$x \equiv 4 \pmod{11}$$

3. Write a program (in C or Python) to solve CRT.

Homework.

Homework.

4. Complete the proof that it is an isomorphism from $\mathbb{Z}_n^*$ to $\mathbb{Z}_p^* \times \mathbb{Z}_q^*$.
5. Let $p = 5$ and $q = 7$, $n = p \cdot q$. Please explicitly give the correspondence between $\mathbb{Z}_n^*$ and $\mathbb{Z}_p^* \times \mathbb{Z}_q^*$. Hint: Programming is permitted.

Homework.

Homework.

6. 求解 $x^{113} \equiv 15 \pmod{221}$ [提示：这是一道似曾相识的习题，然而中国剩余定理能保证你顺利完成目标。]